

Implementation Calculus of wasm4pm

Exact Formulæ as Implemented in Rust Source

wasm4pm Platform

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Notation

All symbols correspond directly to Rust variable names in the source. Types are given in parentheses where disambiguation matters. File paths are cited at each section head. No background theory is included; every formula is the verbatim implementation.

1 Reinforcement Learning

1.1 Reward Function

File: wasm4pm/src/rl_orchestrator.rs, lines 134–174.

Signature: `compute_reward(prev_health, curr_health, spc_alert_count, guard_pass, circuit_allowed, latency_budget_exceeded)`

$$R = \underbrace{\begin{cases} +1.0 & \text{if } h' < h \\ +0.2 & \text{if } h' = h \\ -1.0 & \text{if } h' > h \end{cases}}_{\text{health component}} - \min(\text{spc_alert_count} \times 0.3, 1.5) + \begin{cases} +0.1 & \text{if } \text{guard_pass} \wedge \text{circuit_allowed} \\ -0.5 & \text{otherwise} \end{cases} - 0.3 \cdot 1 \quad (1)$$

where $h = \text{prev_health} \in \{0, \dots, 4\}$ (u8) and $h' = \text{curr_health} \in \{0, \dots, 4\}$ (u8). Reward bounds: $R \in [-5.3, +1.1]$.

1.2 Health State

File: wasm4pm/src/rl_orchestrator.rs, lines 108–120.

Variables: `event_count` (u64), `trace_count` (u64), `unique_activities` (u64).

$$h = \begin{cases} 4 & \text{if } \text{event_count} = 0 \vee \text{unique_activities} = 0 \\ 3 & \text{if } \text{trace_count} = 0 \\ 2 & \text{if } \text{unique_activities} \leq 1 \wedge \text{event_count} < 5 \\ 1 & \text{if } \text{unique_activities} \leq 2 \wedge \text{event_count} < 20 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

1.3 Q-Learning (Off-Policy TD)

File: wasm4pm/src/reinforcement.rs, lines 160–189.

Variables: $\alpha = \text{learning_rate}$ (f32, default 0.1), $\gamma = \text{discount_factor}$ (f32, default 0.99).

$$\text{max_next_q} = \begin{cases} 0 & \text{if done} \\ \max_{a'} Q(s', a') & \text{otherwise} \end{cases} \quad (3)$$

$$\boxed{Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \text{max_next_q} - Q(s, a) \right]} \quad (4)$$

1.4 SARSA (On-Policy TD)

File: wasm4pm/src/reinforcement.rs, lines 341–362.

Identical structure to Q-Learning except $\text{next_q} = Q(s', a')$ where a' is the *actual* next action taken.

$$\boxed{Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma Q(s', a') - Q(s, a) \right]} \quad (5)$$

1.5 Double Q-Learning (Bias-Corrected)

File: wasm4pm/src/reinforcement.rs, lines 567–618.

Variables: q_a, q_b (separate tables), bootstrap (f32), update_a_first (bool).

A Bernoulli coin-flip with $p = 0.5$ selects which table to update:

$$\begin{aligned} \text{With prob. 0.5: } \text{bootstrap} &= Q_B(s', \arg \max_{a'} Q_A(s', a')) \\ Q_A(s, a) &\leftarrow Q_A(s, a) + \alpha \left[r + \gamma \text{bootstrap} - Q_A(s, a) \right] \end{aligned} \quad (6)$$

$$\begin{aligned} \text{With prob. 0.5: } \text{bootstrap} &= Q_A(s', \arg \max_{a'} Q_B(s', a')) \\ Q_B(s, a) &\leftarrow Q_B(s, a) + \alpha \left[r + \gamma \text{bootstrap} - Q_B(s, a) \right] \end{aligned} \quad (7)$$

1.6 Expected SARSA

File: wasm4pm/src/reinforcement.rs, lines 783–814.

Variables: expected_next (f32), max_q, sum_q, eps = ε , n = $|A|$.

$$\text{expected_next} = \begin{cases} 0 & \text{if done} \\ (1 - \varepsilon) \max_{a'} Q(s', a') + \frac{\varepsilon}{|A|} \sum_{a'} Q(s', a') & \text{otherwise} \end{cases} \quad (8)$$

$$\boxed{Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \mathbb{E}_\pi[Q(s', a')] - Q(s, a) \right]} \quad (9)$$

1.7 REINFORCE (Monte Carlo Policy Gradient)

File: wasm4pm/src/reinforcement.rs, lines 995–1048.

Variables: theta (HashMap<S, Vec<f32>>), g_t (f32), returns (Vec<f32>), pi_j (f32).

Return computation (backward pass):

$$G_t = r_t + \gamma G_{t+1}, \quad G_T = 0 \quad (10)$$

Softmax policy:

$$\pi(a \mid s) = \frac{\exp(\theta_{s,a} - \max_{a''} \theta_{s,a''})}{\sum_j \exp(\theta_{s,j} - \max_{a''} \theta_{s,a''})} \quad (11)$$

Weight update (log-softmax gradient):

$$\theta_{s,a} \leftarrow \theta_{s,a} + \alpha G_t (1 - \pi(a \mid s)) \quad (12)$$

$$\theta_{s,j \neq a} \leftarrow \theta_{s,j} - \alpha G_t \pi(j \mid s) \quad (13)$$

1.8 LinUCB Contextual Bandit (Sherman–Morrison)

File: wasm4pm/src/ml/linucb.rs, lines 130–254.

Variables: $x \in \mathbb{R}^8$ (feature vector), $W \in \mathbb{R}^{5 \times 8}$ (w), $b \in \mathbb{R}^5$ (b), $A \in \mathbb{R}^{8 \times 8}$ (a), A^{-1} (a_inv), $\alpha_{\text{exp}} = \sqrt{2}$ (alpha), $\alpha_{\text{lr}} = 0.1$ (alpha_lr).

Action selection:

$$\text{ucb_bonus} = \alpha_{\text{exp}} \sqrt{x^\top A^{-1} x} \quad (14)$$

$$a^* = \arg \max_a \left(W_a \cdot x + b_a + \text{ucb_bonus} \right) \quad (15)$$

Weight update:

$$\delta = r - (W_{a^*} \cdot x + b_{a^*}) \quad (16)$$

$$W_{a^*} \leftarrow W_{a^*} + \alpha_{\text{lr}} \delta x \quad (17)$$

$$b_{a^*} \leftarrow b_{a^*} + \alpha_{\text{lr}} \delta \quad (18)$$

Covariance update (rank-1 Sherman–Morrison):

$$u = A^{-1} x \quad (19)$$

$$\text{denom} = 1 + x^\top u \quad (20)$$

$$A^{-1} \leftarrow A^{-1} - \frac{u u^\top}{\text{denom}} \quad (21)$$

$$A \leftarrow A + x x^\top \quad (22)$$

2 Statistical Process Control

2.1 Western Electric Rules

File: wasm4pm/src/spc.rs, lines 78–139.

Variables: value (f64), ucl, lcl, cl (f64), recent (&[ChartData]).

Rule 1: value > ucl $\vee$ value < lcl (Out of Control)

Rule 2: $\bigwedge_{i=0}^8 \text{recent}[i].\text{value} > \text{cl} \vee \bigwedge_{i=0}^8 \text{recent}[i].\text{value} < \text{cl}$ (Shift)

Rule 3: $\bigwedge_{i=0}^4 \text{recent}[i+1].\text{value} > \text{recent}[i].\text{value} \vee \bigwedge_{i=0}^4 \text{recent}[i+1].\text{value} < \text{recent}[i].\text{value}$ (Trend)

2.2 Process Capability Indices

File: wasm4pm/src/spc.rs, lines 187–243.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (23)$$

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}} \quad (24)$$

$$C_p = \frac{\text{USL} - \text{LSL}}{6s} \quad (25)$$

$$C_{pk} = \min\left(\frac{\text{USL} - \bar{x}}{3s}, \frac{\bar{x} - \text{LSL}}{3s}\right) \quad (26)$$

$$z_U = \frac{\text{USL} - \bar{x}}{s}, \quad z_L = \frac{\text{LSL} - \bar{x}}{s} \quad (27)$$

$$P_{\text{defective}} = [1 - \Phi(z_U)] + \Phi(z_L) \quad (28)$$

$$\text{DPMO} = P_{\text{defective}} \times 10^6 \quad (29)$$

$$\sigma_{\text{level}} = \Phi^{-1}(1 - P_{\text{defective}}) + 1.5 \quad (30)$$

2.3 Normal CDF (Abramowitz & Stegun Approximation)

File: wasm4pm/src/spc.rs, lines 274–286.

Variables: t, d, z (f64).

$$t = \frac{1}{1 + 0.2316419 |z|} \quad (31)$$

$$d = 0.39894228 e^{-z^2/2} \quad (32)$$

$$\tilde{\Phi}(z) = 1 - d \cdot t \left(0.319381530 + t(-0.356563782 + t(1.781477937 + t(-1.821255978 + 1.330274429 t))) \right) \quad (33)$$

$$\Phi(z) = \begin{cases} \tilde{\Phi}(z) & z > 0 \\ 1 - \tilde{\Phi}(-z) & z \leq 0 \end{cases} \quad (34)$$

2.4 EWMA

File: wasm4pm/src/prediction_drift.rs, lines 97–110.

Variables: alpha = $\alpha \in (0, 1]$ (f64), smoothed (Vec<f64>), prev (f64).

$$s_i = \begin{cases} x_0 & i = 0 \\ \alpha x_i + (1 - \alpha) s_{i-1} & i \geq 1 \end{cases} \quad (35)$$

α is clamped to [f64::MIN_POSITIVE, 1.0] before use.

3 Prediction Subsystem

3.1 N-gram Log Probability

File: wasm4pm/src/prediction.rs, lines 116–145.

Variables: log_prob (f64), prefix (Vec<String>), prob (f64).

Let $A = (a_0, a_1, \dots, a_T)$ be an activity sequence and n the n -gram order. For each position i :

$$\text{prefix}_i = (a_{i-n+1}, \dots, a_{i-1}) \quad (\text{context window of length } n-1) \quad (36)$$

$$P(a_i \mid \text{prefix}_i) = \frac{\text{counts}[\text{prefix}_i][a_i]}{\sum_a \text{counts}[\text{prefix}_i][a]} \quad (37)$$

$$\text{log_prob} = \sum_{i=1}^T \ln P(a_i \mid \text{prefix}_i) \quad (38)$$

3.2 Prediction Confidence & Entropy

File: wasm4pm/src/prediction_next_activity.rs, lines 44–74.

Variables: confidence, entropy_val, ent, max_ent (f64).

$$\text{confidence} = p_1 \quad (\text{probability of top-1 candidate}) \quad (39)$$

$$H = - \sum_i p_i \ln p_i \quad (40)$$

$$H_{\max} = \ln |\text{candidates}| \quad (41)$$

$$\text{normalized_entropy} = \begin{cases} H/H_{\max} & H_{\max} > 0 \\ 0 & \text{otherwise} \end{cases} \quad (42)$$

3.3 Beam Search

File: wasm4pm/src/prediction_next_activity.rs, lines 141–191.

Variables: beams (Vec<(Vec<String>, f64)>), new_prob (f64), beam_width.

At each step, the beam set $\mathcal{B}$ evolves as:

$$\mathcal{B}_{t+1} = \text{top-}w \left\{ (\sigma \oplus a, p_\sigma \cdot P(a \mid \sigma)) \mid (\sigma, p_\sigma) \in \mathcal{B}_t, a \in \mathcal{A} \right\} \quad (43)$$

where $w = \text{beam_width}$, $\oplus$ denotes sequence concatenation, and beams are ranked by p_σ descending.

3.4 Weibull Remaining Time

File: wasm4pm/src/prediction_remaining_time.rs, lines 70–112.

Variables: cv (coefficient of variation, f64), k (shape), lambda (scale), t (elapsed), cumulative_hazard, hazard_rate, survival, median_remaining (f64).

Parameter estimation from empirical moments:

$$k = \text{cv}^{-1.086} \quad (44)$$

$$\lambda = \frac{\bar{d}}{\Gamma(1 + 1/k)} \quad (45)$$

where $\bar{d}$ is the empirical mean duration.

Hazard, survival, and prediction at elapsed time t :

$$h(t) = \frac{k}{\lambda} \left(\frac{t}{\lambda} \right)^{k-1} \quad (46)$$

$$S(t) = \exp \left[- \left(\frac{t}{\lambda} \right)^k \right] \quad (47)$$

$$H(t) = \left(\frac{t}{\lambda} \right)^k \quad (48)$$

$$\text{median_remaining} = \lambda (H(t) + \ln 2)^{1/k} - t \quad (49)$$

3.5 Remaining Time Confidence

File: wasm4pm/src/prediction_remaining_time.rs, lines 556–580.

Variables: size_factor, bucket_cv, global_cv, precision_factor, confidence (f64).

$$\text{size_factor} = \frac{\text{count}}{\text{count} + 10} \quad (50)$$

$$\text{precision_factor} = \max \left(0, \min \left(1, 1 - \frac{\text{bucket_cv}}{\text{global_cv}} \right) \right) \quad (51)$$

$$\text{confidence} = \min(0.6 \cdot \text{size_factor} + 0.4 \cdot \text{precision_factor}, 0.99) \quad (52)$$

3.6 Jaccard Drift Distance

File: wasm4pm/src/prediction_drift.rs, lines 82–89.

Variables: a, b (HashSet<String>), union (usize), inter (usize).

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (53)$$

$$d_J(A, B) = 1 - J(A, B), \quad d_J(\emptyset, \emptyset) := 0 \quad (54)$$

4 Conformance Checking

4.1 Token-Based Replay Fitness

File: wasm4pm/src/conformance.rs, lines 195–203.

Variables: c (consumed), p (produced), m (missing), r (remaining) (f64).

$$c = \max(\text{consumed_tokens}, 1) \quad (55)$$

$$p = \max(\text{produced_tokens}, 1) \quad (56)$$

$$f_{\text{trace}} = 0.5 \left(1 - \frac{m}{c} \right) + 0.5 \left(1 - \frac{r}{p} \right) \quad (57)$$

$$f_{\text{trace}} = \text{clamp}(f_{\text{trace}}, 0.0, 1.0) \quad (58)$$

This is a symmetric weighted split of the Van der Aalst (2004) fitness formula, with equal weight on the consumption and production components.

5 Discovery Algorithms

5.1 Genetic Algorithm — Fitness-Proportionate Selection

File: wasm4pm/src/genetic_discovery.rs, lines 367–396.

Variables: `total` (f64), `threshold` (f64), `fitness` per individual.

$$\text{total} = \sum_i f_i \quad (59)$$

$$\text{threshold} = U[0,1) \times \text{total} \quad (60)$$

$$\text{select } j = \min \left\{ j \mid \sum_{i=1}^j f_i \geq \text{threshold} \right\} \quad (61)$$

5.2 Ant Colony Optimisation — Pheromone Update

File: wasm4pm/src/genetic_discovery.rs, lines 450–525.

Variables: `tau` (pheromone), `eta` (heuristic, normalised edge frequency), `alpha`= 1.0, `beta`= 2.0, `evaporation_rate`= 0.1, `q`= 100.0, `deposit` (f64).

Edge selection probability:

$$P(e) \propto \tau_e^\alpha \eta_e^\beta \quad (\alpha = 1, \beta = 2) \quad (62)$$

Evaporation:

$$\tau_e \leftarrow (1 - \rho) \tau_e, \quad \rho = 0.1 \quad (63)$$

Deposit:

$$\text{deposit} = q \cdot f_{\text{ant}} \quad (64)$$

$$\tau_e \leftarrow \tau_e + \text{deposit} \quad \forall e \in \text{ant path} \quad (65)$$

where $q = 100$ and f_{ant} is the fitness of the ant's discovered model.

6 Machine Learning

6.1 Ordinary Least Squares Linear Regression

File: wasm4pm/src/ml/regression.rs, lines 32–108.

Variables: `sum_x`, `sum_y`, `sum_xy`, `sum_xx`, `sum_yy` (f64), `n` (usize).

$$\text{denom}_x = n \cdot \sum x_i^2 - \left(\sum x_i \right)^2 \quad (66)$$

$$\text{denom}_y = n \cdot \sum y_i^2 - \left(\sum y_i \right)^2 \quad (67)$$

$$\text{slope} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\text{denom}_x} \quad (68)$$

$$\text{intercept} = \frac{\sum y_i - \text{slope} \cdot \sum x_i}{n} \quad (69)$$

$$R^2 = \frac{(n \sum x_i y_i - \sum x_i \sum y_i)^2}{\text{denom}_x \cdot \text{denom}_y} \quad (70)$$

6.2 PCA — 2×2 Analytical Eigendecomposition

File: wasm4pm/src/ml/pca.rs, lines 54–149.

Variables: mean_x, mean_y, cov_00, cov_01, cov_11 (f64), lambda1, lambda2, discriminant, explained_variance ([f64;2]).

Covariance matrix:

$$\Sigma = \frac{1}{n-1} \begin{pmatrix} \sum (x_i - \bar{x})^2 & \sum (x_i - \bar{x})(y_i - \bar{y}) \\ \sum (x_i - \bar{x})(y_i - \bar{y}) & \sum (y_i - \bar{y})^2 \end{pmatrix} = \begin{pmatrix} \text{cov_00} & \text{cov_01} \\ \text{cov_01} & \text{cov_11} \end{pmatrix} \quad (71)$$

Eigenvalues (characteristic polynomial):

$$\text{tr}(\Sigma) = \text{cov_00} + \text{cov_11} \quad (72)$$

$$\det(\Sigma) = \text{cov_00} \cdot \text{cov_11} - \text{cov_01}^2 \quad (73)$$

$$\text{discriminant} = \max\left(0, \frac{\text{tr}(\Sigma)^2}{4} - \det(\Sigma)\right) \quad (74)$$

$$\lambda_{1,2} = \frac{\text{tr}(\Sigma)}{2} \pm \sqrt{\text{discriminant}} \quad (75)$$

Explained variance:

$$v_k = \frac{\lambda_k}{\lambda_1 + \lambda_2}, \quad k \in \{1, 2\} \quad (76)$$

7 Summary of Constants

Symbol	Value	Location
α (Q-Learning)	0.1	reinforcement.rs
γ (discount)	0.99	reinforcement.rs
ε (exploration)	0.1	reinforcement.rs
α_{exp} (LinUCB)	$\sqrt{2}$	ml/linucb.rs
α_{lr} (LinUCB)	0.1	ml/linucb.rs
λ_{reg} (LinUCB)	1.0	ml/linucb.rs
SPC Rule 2 window	9	spc.rs
SPC Rule 3 window	6	spc.rs
SPC alert penalty	0.3 per alert	rl_orchestrator.rs
SPC cap	1.5	rl_orchestrator.rs
ACO α	1.0	genetic_discovery.rs
ACO β	2.0	genetic_discovery.rs
ACO evaporation ρ	0.1	genetic_discovery.rs
ACO deposit q	100.0	genetic_discovery.rs
Weibull shape exponent	-1.086	prediction_remaining_time.rs
Confidence size saturation	10	prediction_remaining_time.rs
Confidence floor	0.6 (size), 0.4 (precision)	prediction_remaining_time.rs
Normal CDF p_1	0.2316419	spc.rs
Normal CDF p_2	0.319381530	spc.rs

Table 1: Hardcoded constants in the wasm4pm Rust implementation.